IEEE Access
Multidisciplinary ⋮ Rapid Review ⋮ Open Access Journal

Received 2 September 2024, accepted 20 September 2024, date of publication 30 September 2024, date of current version 9 January 2025.

Digital Object Identifier 10.1109/ACCESS.2024.3470319

RESEARCH ARTICLE

Fast and High-Precision Human Fall Detection Using Improved YOLOv8 Model

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ABSTRACT Human falls refer to a sudden, accidental, unintentional, involuntary, and traumatizing action of a person losing stability and ending up lying down (often on the ground). These incidents of collapsing, tripping, crashing, tumbling, falling, etc. cause major harm and injuries to the elderly in our society. According to WHO, it is the second-leading cause of death worldwide. Therefore, having a fast and high-precision fall detection system operating in real time is crucial. It will ensure quick assistance and early treatment are provided to the impacted elderly individuals. Recently, Deep Learning based systems have shown promising results for detecting changes in postures and promptly responding to them in real-time. The You Only Look Once (YOLO) models have been widely used previously for designing and implementing human fall detection systems. This research presents an improved YOLOv8 model for fast and highly accurate human fall detection. The improvements include reducing the number of layers in the backbone of YOLOv8 and incorporating the attention mechanism in the head of the network. For training and evaluation, the CAUCAFall dataset is used. The original YOLOv8 and improved YOLOv8 are evaluated on the same dataset, achieving mAP of 0.995. Also, the training time of the improved YOLOv8 is 0.457 hours, faster than other YOLO models. Therefore, the presented model is a practical approach for human fall detection and can serve in public places for safeguarding the elderly of our society against potential injuries.

INDEX TERMS Human fall detection, deep learning, YOLOv8, attention mechanism, lightweight network, parameter reduction.

I. INTRODUCTION

In elderly humans, typically above the age of 60, health and well-being become increasingly significant concerns. Among the myriad challenges faced by the elderly population, the risk of human falls stands out as a critical issue with far-reaching consequences. WHO estimates that the number of people over 60 years of age will double by 2050 [1], making it graver and more serious.

A fall is an unpredictable and traumatizing event where a person finds himself lying down in a lower position, often on the ground or floor [2]. It can take place at the residence of the person where they are all alone or in public places where they go for some reason. But, the results of a human fall are very serious, including physical injuries, fractures, mental setbacks, loss of self-confidence, and even accidental death. Following a human fall, the person might not be able to call for help or lose consciousness, leading to a delay in timely first-aid and subsequent treatment [3].

According to the World Health Organization (WHO), an estimated 37.3 million falls requiring medical attention occur each year and falls are the second leading cause of death worldwide. Mostly, adults above 60 years of age suffer the highest number of fatal falls. Therefore, prevention strategies for both residential and public places are of paramount significance [4]. There is an urgent need to tackle the issue of identifying human falls with the help of the latest technology to ensure the elderly population doesn't remain unattended and gets a quick response after a fall [5]. Recently, human fall detection has gained attention in e-healthcare. It refers to identifying an elderly person when he has tripped, stumbled, collapsed, crashed, kneeled,

The associate editor coordinating the review of this manuscript and approving it for publication was Zahid Akhtar.

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or in the way found positioned on the lower surface [6]. Traditional methods of fall detection, such as wearable sensors or motion detectors, have limitations in accuracy, reliability, and scalability. These methods often suffer from high false alarm rates, limited coverage, and discomfort for the wearer [7]. Consequently, there is a growing interest in leveraging advanced technology, particularly deep learning algorithms, to improve fall detection systems. The YOLOv8 algorithm is known for its real-time processing speed and high accuracy, making it an attractive and ideal choice for fall detection [8]. However, the performance of these systems is limited by several challenges including variability in fall patterns, diverse environmental conditions, occlusions, and the need for real-time processing with low computational resources [9].

In this research, we present a fast and high-precision fall detection system that can be used to help the elderly after a fall in residential, occupational, or public places. The system is based on the YOLOv8 algorithm, which is popular for object detection tasks. In this approach, the existing YOLOv8 algorithm has been improved to enhance its accuracy and speed up its result delivery. We list our contributions to the field as follows:

1) Proposed an improved backbone by Reducing the number of layers in the original backbone of YOLOv8, in doing so we reduce the system complexity and consequentially, improve the model speed.
2) Incorporating the Ghost Bottleneck attention modules [10] in the head to suppress irrelevant information and improve the quality of detection.

The rest of this paper is organized as follows. Sec. II reviews the previous related work for human fall detection. Sec. III introduces the vision-based approaches, attention mechanism, and YOLOv8 algorithm. Sec. IV presents the proposed method of human fall detection, the dataset, and the hardware and software requirements. Finally, Sec. V presents the results of the work, comparing it with other previous studies in the same direction. Lastly, Sec. VI concludes with the findings of the research, acknowledging the limitations and suggesting future works.

II. LITERATURE REVIEW

Computer-aided human fall detection has garnered significant attention from researchers and healthcare professionals due to its crucial role in preventing injuries and improving the quality of life for the elderly population. Developing advanced fall detection systems for the elderly can help them in several ways. These include leading a normal life, following their routines independently, reducing their healthcare expenditures, and preventing them from isolation. Altogether, it can boost their physical and mental well-being, raising their quality of life. This section discusses existing literature on human fall detection methods, including computer vision techniques such as YOLOv8, reduction in the number of layers in deep networks, and incorporation of attention mechanisms in architectures. This review aims to identify trends, gaps, and future directions for advancing human fall detection technology.

A. FALL DETECTION WITH YOLO

Recently, the YOLO family of models has been exploited for human fall detection. The increased adoption can be attributed to YOLO's popularity for real-time object detection with high accuracy and its ability to process images in a single pass through the network. For example, Lu and Chu [11] developed an image-based fall detection system for the elderly, when they are trying to sit or get up from a chair, using YOLOv3 for detection and Camshift for tracking. Long et al. [12] leveraged the YOLOv3 algorithm for human fall detection (under daylight conditions). Kamble et al. [13] also utilized the YOLO algorithm width-height (wh) method to detect falls, especially at night (low-light conditions). When humans were recognized in a video frame, the bounding box was drawn around them, and width and height were calculated. If width was greater than height for several frames, a fall event was detected.

Advancing further, Sarıçayır et al. [3] implemented two different architectural configurations of the YOLO framework - YOLOv7 and YOLOv8 for the human fall detection task. The YOLOv7 algorithm was more accurate but slower in delivering results when compared to YOLOv8. Similarly, Ke et al. [14] tested the YOLOv5 variants on the CAUCAFall dataset and found YOLOv5x to be the most accurate. But it was slower than YOLOv5s, which was slightly behind in accuracy but offered good speed for real-time fall detection. Several studies have explored reducing the number of layers and/or using ghost convolution techniques to minimize the computational overload and complexity of deep networks like YOLO. For instance, Kan et al. [15] designed CGNS-YOLO for fall detection by reconfiguring the neck network of YOLO5 with the integration of the GSConv module and GDCN module. They also used a normalization-based attention module (NAM) to refine the method's precision. Zheng et al. [16] presented the FDT-YOLO algorithm that enhanced fall detection by modifying the YOLOv8 framework, replacing the C2f module with the FasterNet module for better feature reuse and reduced computational complexity. It also integrates a deformable convolution module and triplet attention mechanism, leading to improved detection accuracy and robustness. Luo et al. [17] developed YOLOv5s-GCC by replacing the original Conv and C3 structures with GhostConv and C3GhostV2 in the backbone of YOLOv5s and using a lightweight upsampling operator CARAFE for improved feature fusion. Additionally, they incorporated the CBAM attention mechanism in the deepest neck layer. Attention mechanisms have gained prominence in enhancing fall detection systems by allowing models to focus on critical features. For example, Zhao et al. [18] introduced YOLOv7-fall, by adopting a novel attention module, SDI, based on Coordinate Attention and Shuffle Attention techniques. Additionally, they used ghost convolutions in the head to reduce parameters and computational complexity.

TABLE 1. Summery of related work on fall detection with YOLO.

Reference	Methodology	Dataset	Metrics	Result	Limitations
Lu and Chu [11]	Deploy YOLOv3 for detection and Camshift for tracking	Simulated fall scenarios, Youtube and Giphy videos	Accuracy	0.9596	Inaccuracies in YOLOv3 and Camshift tracking can cause fall detection false negatives with challenges in complex movements or occlusions.
Long et al. [12]	Leveraged YOLOv3 under daylight conditions	-	Accuracy	0.9200	Reliance on clear visibility may limit effectiveness in poorly lit environments, making detection performance vulnerable to varying lighting conditions
Kamble et al. [13]	Applied YOLO with width-height method to detect falls	UCF-101	Accuracy	0.9316	Limited to width-height criteria which may not met due to positional variations. Furthermore, dynamic or obstructive scenarios may increase false negatives.
Sarıçayır et al. [3]	YOLOv7 and YOLOv8 frameworks	Fallen New Version Dataset	Accuracy	0.8100 and 0.7800; respectively	YOLOv7 may have delays in processing, impacting real-time applications, and YOLOv8's faster processing might sacrifice some accuracy, leading to possible false negatives.
Ke et al. [14]	Tested all YOLOv5 variants, identifying YOLOv5x as the most accurate.	CAUCAFall	Precision	0.8220	YOLOv5x suffers from computational cost speed, which hinders its application in real-time detection.
Kan et al. [15]	Designed CGNS-YOLO with custom modules for precision improvement	URFall, Multicam and Le2i Fall Detection dataset	mAP	0.9130	Reconfiguration may introduce instability in performance due to reliance on the new modules. Furthermore, complexity introduced by multiple modules may hinder interpretability and debugging.
Zheng et al. [16]	Developed FDT-YOLO by enhancing the YOLOv8 framework with new modules for feature reuse.	Le2i , Multicam, Fall detection datasets, and some real-world images obtained from the internet	mAP	0.9620	Suffers from computational resources, potentially hindering speed. Furthermore, the effectiveness of new modules in diverse environments still needs validation.
Luo et al. [17]	Improved YOLOv5s-GCC	URFall, Fall Detection, and Multicam Datasets	mAP	0.9190	limited by generalize well across different datasets, affecting performance in varied scenarios with inconsistencies in feature extraction efficacy.
Zhao et al. [18]	YOLOv7-fall based attention module	URFall, Multicam, Fall Detection datasets, and photos from internet	mAP	0.9500	focus on specific features may overlook other important contextual cues, leading to missed detections. Furthermore, complexity from added modules can slow down inference times in resource-constrained environments.
Sha et al. [19]	YOLOv7-fall based attention module	COCO, URFall, UMAFall, and Youtube videos	mAP	0.9180	compromise detection accuracy in challenging conditions, and effective inference may need careful tuning, which could add to model complexity.
Chen et al. [20]	Enhanced YOLOv5 with an Efficient Channel Attention network	Public dataset	mAP	0.8829	increased model size and complexity, potentially hindering real-time applications. Furthermore, the precision might vary based on the scale of fall events presented.
Chen et al. [21]	Improved YOLOv5s	. URFall	mAP	0.9720	Removing feature layers can lead to loss of critical information regarding smaller objects, affecting overall detection capabilities. Furthermore, the model may require extensive retraining for optimal performance following architecture modification.
Khan et al. [22]	Integrated a focus module in improved YOLOV8S model to optimize feature extraction	Diverse FALL10500	-	-	Increased in computational load, impacting real-time performance. Furthermore, the dependencies on certain features may limit adaptability in diverse scenarios.
Qin et al. [23]	Enhanced YOLOv8S	URFall, Fall Detection Dataset, and internet images	mAP	0.8870	Difficulties in consistent performance across different lighting and environmental conditions.

Sha et al. [19] incorporated a global-local attention module alongside channel-wise dropout methods in the convolutional neural network. While attention helped to enhance accuracy, the dropout boosted speed and reduced computational cost and model size. Chen et al. [20] enhanced the YOLOv5.

TABLE 2. Summery of previous works for fall detection with other approaches.

Reference	Methodology	Dataset	Metrics	Result	Limitations
McCall et al. [24]	transfer learning on a transformer model trained on 2D poses. Fine-tuned last layers after pre-training on a general action dataset	CAUCAFall	Accuracy	97.95%	Relies on the initial dataset's quality and diversity, which may not fully capture fall scenarios. Furthermore, limited adaptability to varied real-world conditions.
Bansal et al. [25]	Fine-tuned a Faster R-CNN model using COCO weights	CAUCAFall	Accuracy	93.02%	Performance may degrade in environments significantly different from the COCO dataset's context. Furthermore, potential overfitting to COCO's features rather than fall-specific ones.
Nagaj et al. [26]	Segmentation with pixel transformations based ResNet-18, EfficientNetV2-S, and Swin-Small models for detection	CAUCAFall, KULeuven, and URFall	F1-Score	0.8730	Inaccurate segmentation can result in false negatives. Furthermore, specialized pixel transformations might not generalize well across different environments. Moreover, high reliance on computational resources due to complex models used.
Alanazi et al. [27]	Automated vision-based system utilizing a 3D multi-stream CNN model	Le2i	Accuracy	0.9944	Suffers from low generalizing with minimum system's effectiveness in various real-world scenarios, remains to be thoroughly evaluated beyond the dataset.

Architecture by incorporating an improved Efficient Channel Attention (ECA) network. They also integrated average pooling layers within a Spatial Pyramid Pooling (SPP) network to facilitate multi-scale feature extraction. The work could accurately detect fall events that may vary in scale and pose subtlety. Similarly, Chen et al. [21] improved YOLOv5s for fall detection by using asymmetric convolution blocks (ACB) and spatial attention mechanisms in the backbone network. They also removed the feature layer for small targets to improve the focus on fall detection. Khan et al. [22] integrated a focus module in the backbone to optimize feature extraction and Convolutional Block Attention Modules (CBAMs) at reasonable places into an improved YOLOV8S model. Likewise, Qin et al. [23] also improved the YOLOv8S model by replacing C2f with C2Dv3 modules in the backbone and adding a DyHead block in the neck layer for unifying multiple attention operations, as shown in Table 1.

B. FALL DETECTION WITH OTHER APPROACHES

Deep neural network architecture-based methods have been used for fall detection. Concepts such as transfer learning and transformers have been employed for approaches not involving YOLO. Recently, McCall et al. [24] leveraged transfer learning for fall detection and trained a transformer model on 2D poses. They fine-tuned only the last layers of the transformer after training it on a large general action dataset. Similarly, Bansal et al. [25] fine-tuned a Faster R-CNN model with COCO weights for fall detection. Nagaj et al. [26] created a segmentation pipeline for bifurcating the individuals and objects in the image to detect falls. They applied pixel-based transformations to the segmented objects (Gaussian blur) and employed models like ResNet-18, EfficientNetV2-S, and Swin-Small for detection. The work reduced false positives to a large extent. Alanazi and Muhammad [27] introduced an automated vision-based system for detecting falls using 3D multi-stream CNN model and generating instant alerts upon detection of a fall, as shown in Table 2.

III. RELEVANT CONCEPTS

The proposed fall detection method integrates several key concepts. Firstly, it is based on the YOLOv8 algorithm. Secondly, it incorporates the Ghost Bottleneck attention modules in YOLOv8. Lastly, it reduces the number of layers in YOLOv8. This section provides a brief explanation of each of these concepts.

A. PRINCIPLES OF YOLOv8 ALGORITHM

YOLOv8 is the latest version of the You Only Look Once (YOLO) family of object detection algorithms, known for its real-time performance and high accuracy. YOLOv8 builds on the strengths of its predecessors while introducing improvements in speed, accuracy, and efficiency. It employs a single neural network to predict bounding boxes and class probabilities directly from full images in one evaluation, making it highly suitable for real-time applications like fall detection [28]. There are 4 parts in the YOLOv8 architecture - backbone, neck, head, and loss. The backbone extracts essential features from the input images. It employs the Cross Stage Partial (CSP) architecture to split the feature maps into two parts. The convolutional layers process the first part. Then, the backbone concatenates the second part with the output of the first part. It also contains the C2f module which combines the C3 module and the concept of ELAN. This facilitates a more robust gradient information flow throughout the backbone [29]. The YOLOv8 neck enhances features from the backbone. It prepares them for the detection head. It uses the Feature Pyramid Network (FPN) and Path Aggregation Network (PAN) for multi-scale feature fusion.

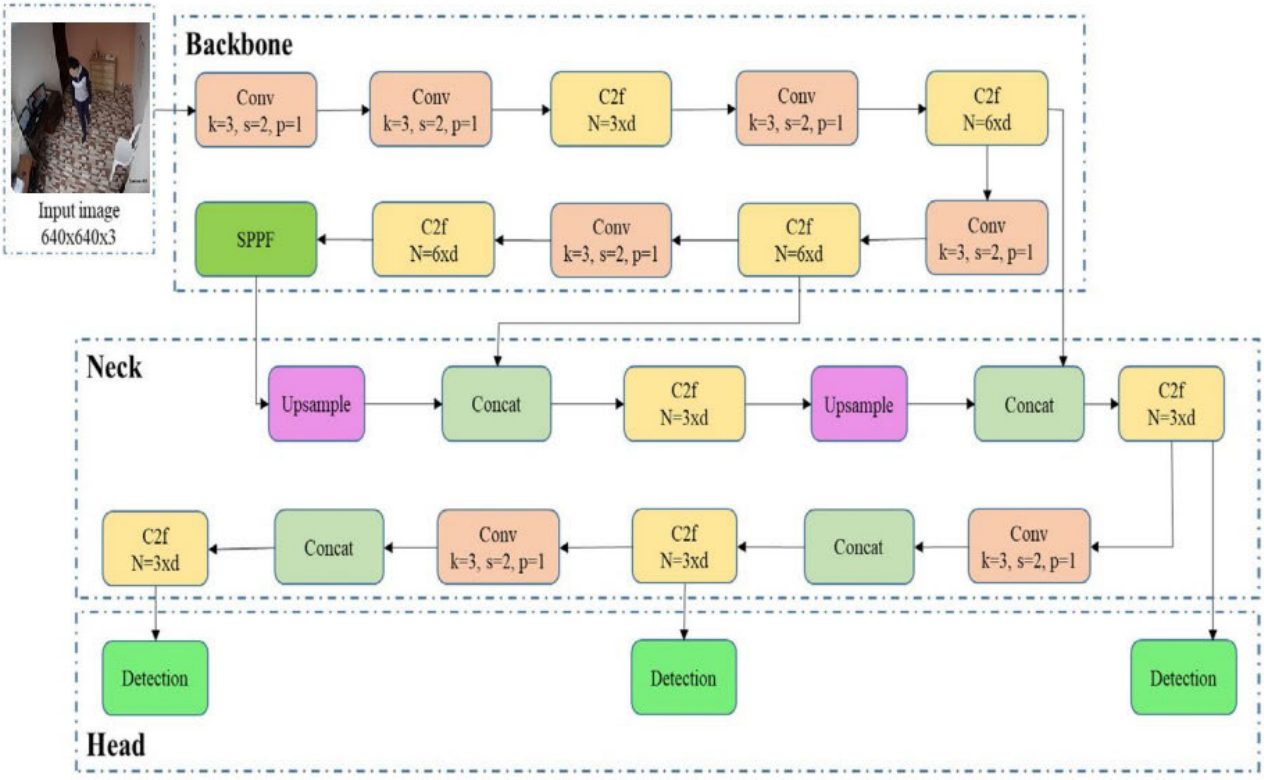


FIGURE 1. Architecture of YOLOv8 algorithm model.

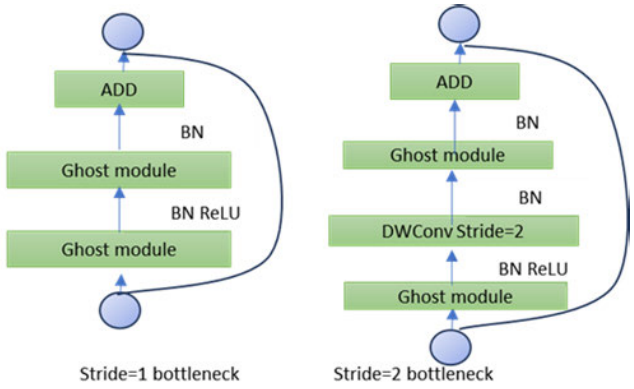


FIGURE 2. Ghost bottleneck attention modules.

Multi-scale feature fusion means combining features from different layers of the backbone. Hence, the location information is preserved as the network gets deeper [30]. The head is where the actual detection happens. It consists of several convolutional layers that take the enhanced features from the neck and predict bounding boxes, object classes, and confidence scores [31]. In YOLOv8, the head is typically decoupled. It means that the tasks of localization and classification are split in the head. They are handled separately within the detection head. For each task, the optimization is done individually. During optimization, different aspects of the learned features are leveraged [32]. Thus, splitting these tasks can improve the accuracy and efficiency of both predictions. The loss function in YOLOv8 is a critical component that guides the training process. It typically combines multiple loss components - classification loss, localization loss, and confidence loss [33]. The architecture of the YOLOv8 algorithm is shown in Fig. 1.

B. GHOST BOTTLENECK ATTENTION MODULES

The Ghost Bottleneck attention modules are an advanced feature that help to enhance the performance of deep convolutional neural networks. These modules leverage the Ghost module concept, which generates more feature maps from inexpensive operations, significantly reducing the computational cost. The attention mechanism within these modules helps the network focus on the most relevant parts of the input data, improving detection accuracy by suppressing irrelevant information and emphasizing critical features [34]. The key idea behind ghost convolutions (GhostConv) is generating more feature maps from fewer convolutional operations (ghosting the features). First, the convolution operation is applied to generate intrinsic feature maps. Then, the sum in expensive linear operations is applied to the intrinsic feature maps to generate additional feature maps, known as ghost feature maps. Lastly, the internship future maps are combined with the ghost future maps to get the final output. In this way the ghost

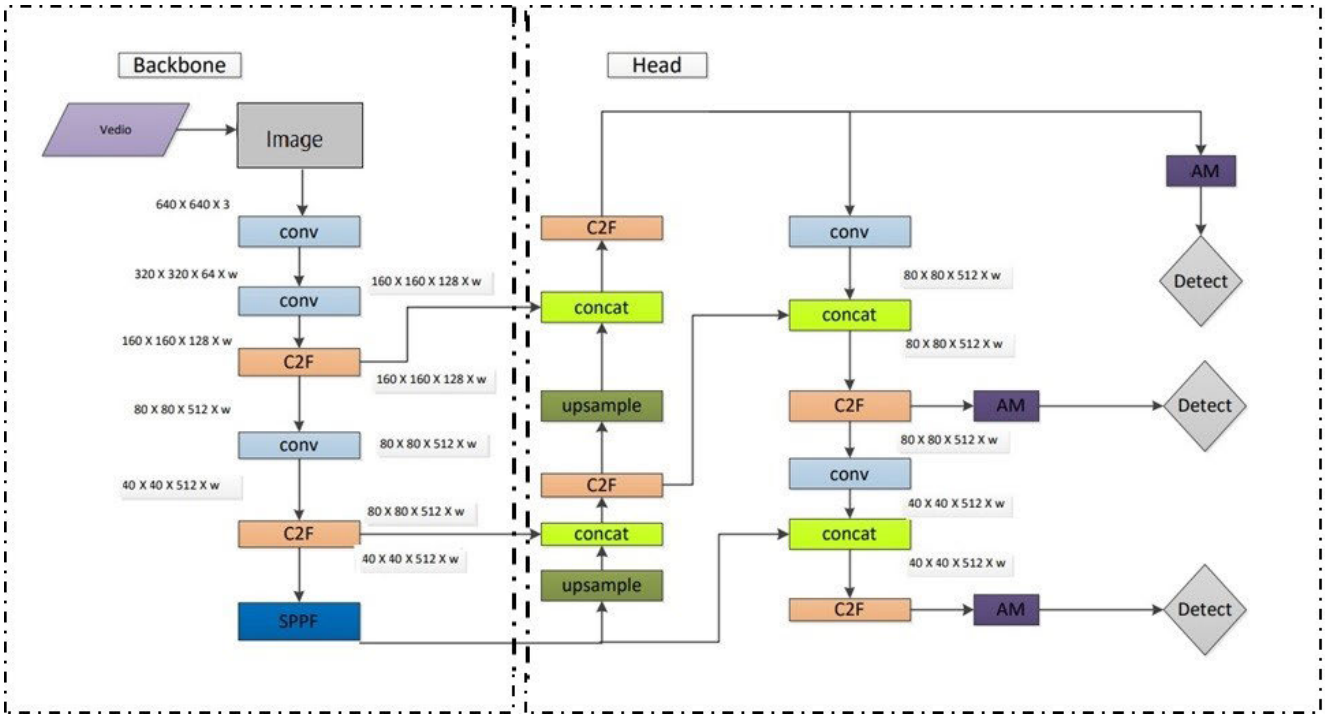


FIGURE 3. The architecture of improved YOLOv8 model.

convolutions can achieve a similar representation to the standard convolutions but with significantly fear parameters and computational requirements [10]. C3Ghost modules are an advanced component that integrates ghost convolutions into the YOLO architecture. The "C3" refers to the Cross Stage Partial Network (CSPNet) which is commonly used in YOLO models to enhance feature extraction and reduce computational load. The C3Ghost modules work by splitting the input into two parts, with first part being processed by the ghost convolution layers and the second part by-passing them [35]. Figure 2 presents a schematic of the Ghost Bottleneck Attention Modules incorporated into the convolutional layer and C3 layer is presented. The ghost convolutions (GhostConv) generate the ghost features and merge them with the intrinsic features. Next, the output of the ghost convolution is concatenated with the bypassed input. Afterwards, the concatenated features pass through a bottleneck design for further refinement and reduction in the dimensionality of the feature maps [36]. Lastly, the refined features are passed through the next layer of the network. In this way, the C3Ghost modules enhance the efficiency of deep learning models making them faster and more suitable for real-time applications without compromising the accuracy [37].

C. REDUCTION IN NUMBER OF LAYERS

The most common way to make CNNs lighter without hurting performance is to cut the number of layers. A strategic decrease in layers enhances inference speed and lowers computational requirements. This optimization is crucial for the model's use on devices with limited processing power. Also, it's necessary for ensuring the real-time performance of AI models [38].

IV. PROPOSED METHOD

The proposed method employs an improved YOLOv8 algorithm to detect incidents such as falling, tripping, collapsing, slipping, tumbling, or in a way positioned on a low level close to the ground, particularly involving elderly individuals. Two improvements are made in the baseline YOLOv8 algorithm for the proposed method. These improvements are guided by the aim of lightweighting, speeding up detection, and reducing misclassifications. First, the number of convolutional layers in the backbone network is reduced. And second, the attention mechanism is incorporated into the head of the network. The architecture of the improved YOLOv8 algorithm is illustrated in Figure 3. Steps involved in improving, training, and evaluating the model are discussed in respective sub-sections.

A. REDUCING LAYERS IN THE BACKBONE NETWORK

The backbone network in YOLOv8 is responsible for extracting essential features from the input images. It processes the images through multiple layers to identify and encode various patterns, textures, and object representations, which are crucial for accurate object detection in the subsequent stages of the algorithm. However, the presence of many convolutional layers in the backbone makes it computationally

intensive and slow, leading to increased processing time and higher resource consumption. Additionally, a very deep backbone might cause vanishing gradient issues, making it harder to train the network effectively and potentially leading to diminishing returns in performance improvements. Therefore, in the proposed method, the number of convolutional layers in the backbone network is reduced. Originally, the YOLOv8 algorithm contains 5 Conv and 4 C2f layers in the backbone. In this work, the backbone is modified to include 4 Conv and 2 C2f layers. This reduction will help to decrease computational complexity, improve processing speed, and reduce resource consumption while maintaining the quality of feature extraction for accurate detection. Figure 3 illustrates the original backbone of YOLOv8 and the modified backbone architecture.



FIGURE 4. Image samples of the CAUCAFall dataset.

B. INCORPORATING ATTENTION MECHANISM IN THE HEAD NETWORK

The head of the YOLOv8 algorithm is responsible for processing the features extracted by the backbone network and generating predictions for object detection. It typically consists of several convolutional layers followed by detection layers that predict bounding boxes, confidence scores, and class probabilities for objects in the input image. These predictions are then used to identify and localize objects of interest within the image. But, one of the major limitations of the head is that it can struggle to localize objects in cluttered or densely populated scenes. It can lead to missed detections or inaccurate localization of objects. Besides, the YOLOv8 head may struggle to detect objects with significant variations in scale or aspect ratio within the same image. These limitations can impact the algorithm's performance in scenarios where precise object detection is crucial, such as in fall detection for the elderly. In this work, we've integrated the attention mechanism module into the YOLOv8 head network, as illustrated in Figure 3. In the original YOLOv8 algorithm, the detection process is initiated by the output from 3 C2F layers. In the improved version, the attention module is positioned after these 3 C2F layers, just before the detection phase. This means that attention is applied to the output of the 3 C2F layers. As a result, irrelevant and unnecessary information at large scales is suppressed, leading to an enhancement in the quality of detection. By incorporating these two improvements in the baseline YOLOv8 algorithm, great optimization in the speed and performance can be achieved. The overall architecture of the proposed method is shown in Fig. 3.

C. EVALUATING THE IMPROVED YOLOv8 MODEL

Two key assessments will evaluate the proposed fall detection method. The first assessment focuses on the accuracy of the method. It measures how effectively it detects fall events among elderly individuals. The second assessment examines the detection speed. It evaluates how quickly the method can identify falls. Both assessments are crucial to determine how effective the improvements in YOLOv8 were. For measuring the detection accuracy, mean Average Precision (mAP) is popularly used. It is a critical metric for evaluating the YOLO models. It considers both precision and recall across different classes and intersection over union (IoU) thresholds. So, it provides a comprehensive evaluation of detection accuracy. In this work, mean Average Precision (mAP) will be used to determine the detection accuracy of the improved YOLOv8 model. Also, the training time of the model will be reported to indirectly examine the processing speed. To demonstrate the effectiveness of the proposed method, it will be compared with the baseline YOLOv8 model. A comparison with other versions of YOLO - YOLOv5, YOLOv6, and YOLOv7, will also be made.

V. EXPERIMENT AND RESULTS

A. DATASET AND TRAINING

In this research, the CAUCAFall [39] dataset is used for training and evaluating the proposed method. This dataset is especially prepared for recognizing human falls in an uncontrol home environment. It consists of 100 videos, which represent five different types of falls—forward, backward, left, right, and seated, besides the daily activities of the elderly. The videos were captured at a frame rate of 23 fps and a resolution of 1080 $\times$ 960 pixels, using a camera setup. Additionally, the videos have either one of the two labels - fall or no-fall.

Notably, the dataset has a lot of variation in the data on account of occlusions, changes in lighting, different types of clothes, movements in the background, different textures on the floor, arrangements in the room, variety of angles, different distances from the camera, people of different age, etc. Hence, it is very suitable for developing solutions related to fall detection, especially through YOLO detectors [39]. However, it is important to note that the CAUCAFall dataset was introduced in the year 2022. Therefore, there have not been many studies conducted on this dataset. Yet, it remains a valuable dataset and has the potential to become a benchmark in years to come. It offers researchers the opportunity to

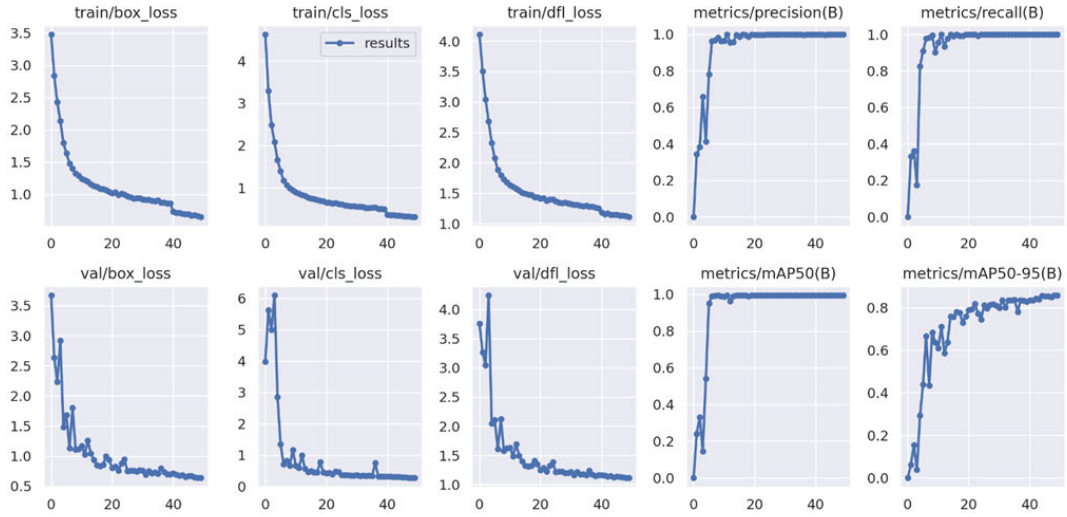


FIGURE 5. Loss curves of improved YOLOv8.

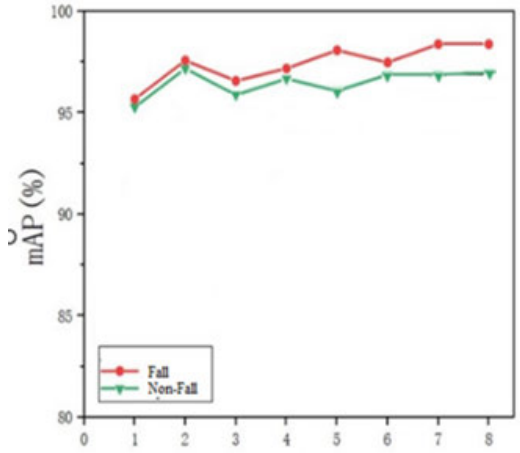


FIGURE 6. The mAP for classes in fall dataset.



FIGURE 7. Detection results of proposed method YOLOv8 on the CAUCAFall datase.

establish benchmarks and standards for future studies. This can foster further research and development in the domain of fall detection and facilitate comparative analysis across different methodologies. Some of the images from the dataset are shown in Fig. 4.

Before training, videos from the CAUCAFall dataset are converted to images and resized to 640 × 640. For the our experiments, the CAUCAFall dataset is divided into three parts train (70%), validation (20%), and test (10%). The train set contains 2297 images and will be used for training the improved YOLOv8 algorithm. The validation set and the test set contain 632 and 327 images, respectively. The validation set will help in tuning the model, while the test set will be used to evaluate the model's final fall detection performance. For model training, the videos will be first converted into images (video frames). Python-based libraries will used for developing the improved YOLOv8 algorithm. For performing the implementation, the Google Colab IDE will be used. Colab is based on Python programming language. In our experiments, we deploy PyTorch environment. It can be configured with two distinct processors to accommodate GPUs or central processing units (CPUs). GPUs typically perform better than CPUs when running PyTorch programs. The improved YOLOv8 model was trained on Google Colab Pro+ using an Nvidia V100 GPU and 58 GB of RAM. The input size for the improved YOLOv8 model is 640 × 640. For training the model 100 epochs are specified with a batch size of 32. Similarly, the other YOLO models were trained with the same settings. The improved YOLOv8 model was trained on the CAUCAFall dataset for 100 epochs. The loss function of the training and validation of the model is presented in

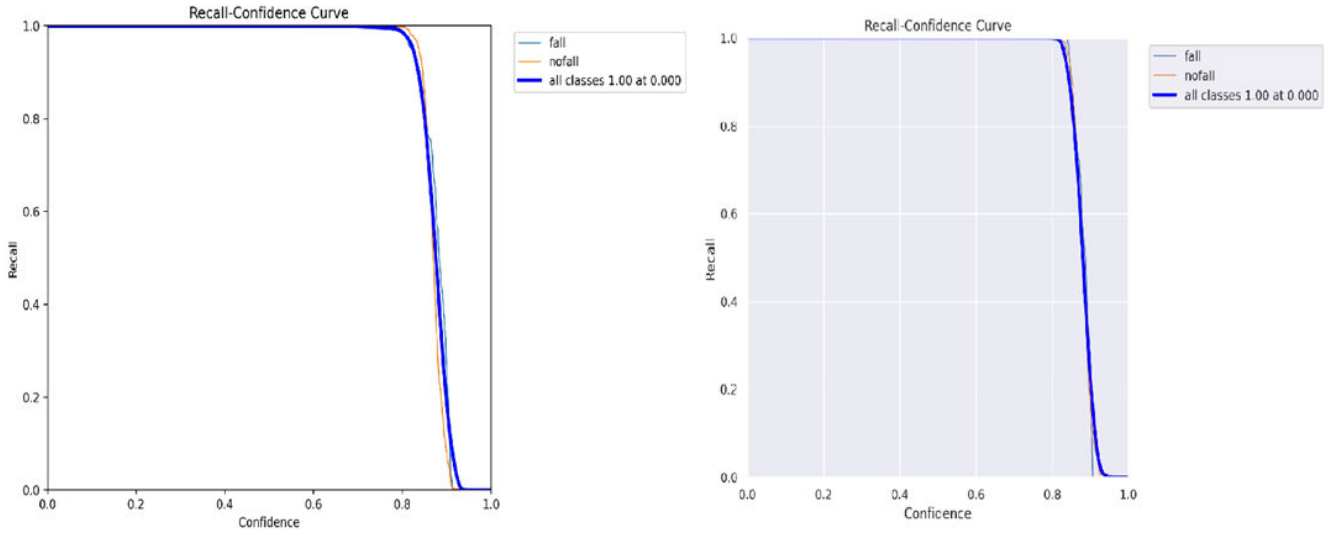


FIGURE 8. Result of fall detection model.

TABLE 3. Implement the model on various type of datasets.

Dataset	Images	Box(P)	R	mAP%
VOC2007 [40]	4952	0.63	0.549	0.606
multi-camera dataset [41]	1208	0.799	0.743	0.80
COCO2014 [42]	5438	0.773	0.536	0.624
LE2I dataset [43]	3540	0.992	0.981	0.980
CAUCAFall [39]	2317	0.999	1	0.995
Computer Vision Project Dataset [44]	1000	0.851	0.781	0.856

Fig. 5, where eight classed are involved in the training as shown in Fig. 6.

B. PERFORMANCE OF THE PROPOSED IMPROVED YOLOv8

The model achieved an impressive detection performance with a mean Average Precision (mAP) of 0.9950. This high accuracy shows that the model performed with high accuracy at identifying fall events among the elderly. The model's training time of 0.4570 hours indicates efficient processing, making it quite reliable for real-time fall monitoring applications. The final validation results of the CAUCAFall dataset can be seen in Fig. 7. Beside that, we present an overall performance of the proposed model with two critical metrics, including recall (sensitivity) and the F1 score, The fall detection model has high capabilities as shown in Fig. 8.

These results shows the fall of four different types, including falls from the front, behind, left and right side. To show the reliability of the proposed model, we included five different public dataset in the experiments, including VOC2007 [40], multi-camera [41], COCO2014 [42], LE2I [43], and Computer Vision Project [44] Datasets. The first four dataset [40], [41], [42], [43] contain traditional fall scenarios and similar to CAUCAFall dataset, while the last dataset [44] contains different types of fall that were not included in the trained CAUCAFall or the other datasets. These falls were extracted from sports competitions and

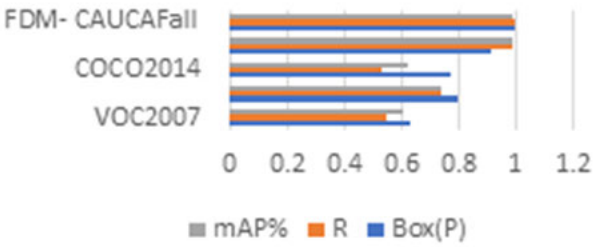


FIGURE 9. Compare of mAP of datasets.

TABLE 4. Comparison with other YOLO models.

Model	layera	parameter	mAP	Training Time (hrs)	Inference time (ms)
YOLOv5	214	702,502	0.830	-	-
YOLOv6	300	343,111	0.828	-	-
YOLOv7	415	37,201,950	0.835	1.043	10.7
YOLOv8	225	3,011,238	0.86	2.9ms	4.9
Proposed method	207	2,155,436	0.995	0.457	5.6

gymsm. There are other cases in which falls are identified, such as falls in the sports field, which is very important in applying sports fitness recognition technology, and it is also an important research topic in the field of artificial intelligence interaction [45], [46], we present the result of the proposed model on Computer Vision Project Dataset in Fig. 10. Moreover, to show the generalization of the proposed model over complex environment, different lighting conditions, and various of fall types, we present an objective comparison of the proposed method for all the involved dataset in Table 3 and Fig. 9.

Actually, we justify the stable performance of the proposed model of the results for two main reasons. Firstly, its lightweight design, comprising 207 layers and 2,155,436 parameters. It enhances its computational efficiency significantly. Secondly, it's the attention mechanism. It levels up the feature extraction process. These results show the improved YOLOv8 model's potential for real-world fall detection scenarios.

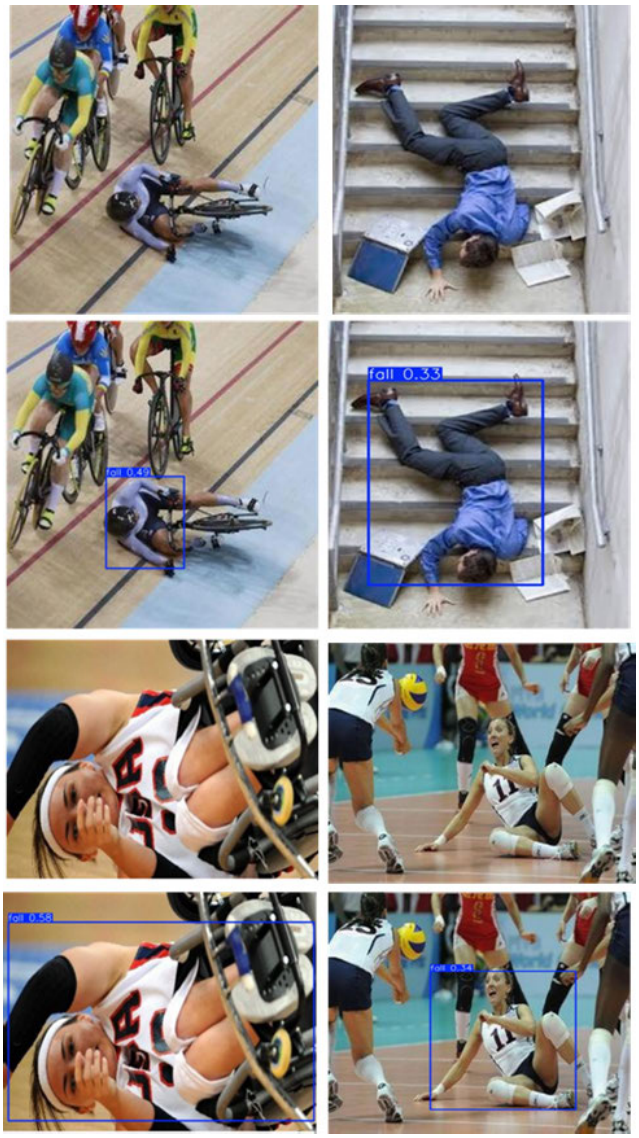


FIGURE 10. Fall computer vision project dataset.

TABLE 5. Comparison with the state-of-the art methods.

Method	Model	mAP Trainable	Parameters
Sarıçayır et al. [3]	YOLOv7	0.81	-
Sarıçayır et al. [3]	YOLOv8	0.78	-
Zheng et al. [16]	Improved YOLOv8	0.962	9979000
Zhao et al. [18]	Improved YOLOv7	0.95	6017744
Tan [47]	EfficientDet	0.89	-
Qin et al. [23]	Improved YOLOv8	0.887	-
Proposed method	Improved YOLOv8	0.995	2155436

C. COMPARISON WITH BASELINE

The proposed model was compared with four versions of the YOLO object detection algorithm, including YOLOv5, YOLOv6, YOLOv7, and YOLOv8. All these models with their original configuration were trained on the CAUCAFall dataset and evaluated for mAP and training time. The objective comparison results are presented in Table 4. Notably, the improved YOLOv8 model uses the fewest layers and achieves the highest mAP of 0.995, showing

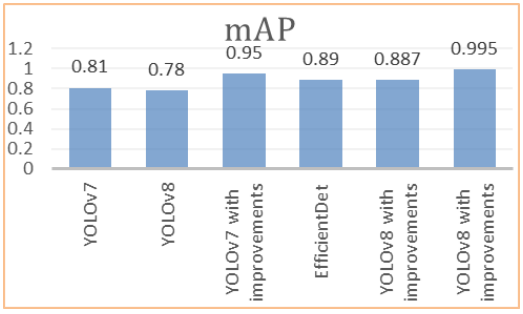


FIGURE 11. mAP comparison of the proposed method with other YOLO models.

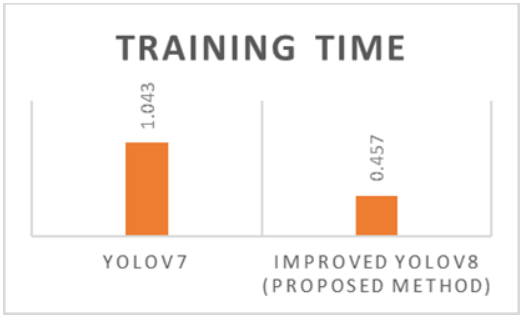


FIGURE 12. Training time comparison of the proposed method with YOLOv7 model.

an improvement of 13.06% in detection performance in comparison to baseline YOLOv7 and YOLOv8 as depicted in Fig. 11. Furthermore, there's a significant 28.42% reduction in model parameters (compared to the baseline YOLOv8 model), highlighting the lightweight nature of the proposed model without compromising accuracy. The training speed of our model is 0.457 hours, significantly faster than the YOLOv7 algorithm, which has a training time of 1.043 hours (as it can be seen in Fig. 12). Hence, it is very suitable for use in real-life applications for detecting human falls among the elderly. These comparisons with the baseline models underscore the efficiency and effectiveness of the proposed fall detection method.

D. COMPARISON WITH THE STATE-OF-THE-ART

The proposed model was also compared with state-of-the-art methods, including Sarıçayır et al. [3], Zhao et al. [18], Qin et al. [23], and Zheng et al. [16]. Sarıçayır et al. [3] proposed a fall detection model with YOLOv7 and YOLOv8, achieving mAP scores of 0.81 and 0.78, respectively. These models were performed without any improvements. To increase the reliability of YOLOv7, Luo et al. [17] introduced enhancements to the YOLOv7 architecture. The authors included SDI attention mechanism, as well as GSConv and GSBottleneck modules. Their work achieved a better a mAP than [3], highlighting the need for further improvements. One of the basic ideas to improve the performance further is the use of more advanced YOLO models. Likewise, [16] incorporated FasterNet module in place of the YOLOv8's C2f module to leverage deformable convolutions

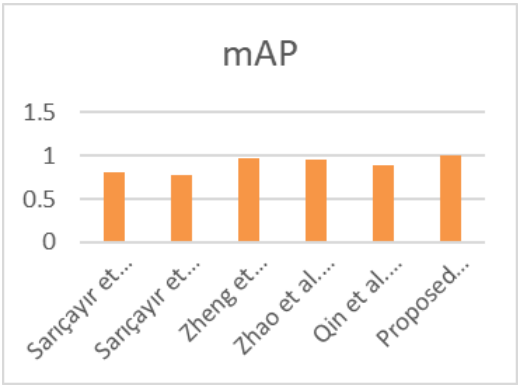


FIGURE 13. Comparison of mAP with state-of-the-art methods.

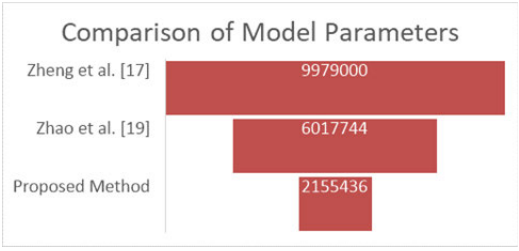


FIGURE 14. Parameters comparison of proposed model with state-of-the-art methods.

and triple attention. An mAP of 0.962 with a total of 9979000 trainable parameters was achieved. Research [22] amended the YOLOv8 by replacing the C2f module in backbone with C2Dv3 module and integrating DyHead block in neck to unify multiple attentional operations. The study got mAP of 0.887. On the other hand, the performance of the proposed improvement on YOLOv8 model achieves a significantly higher mAP of 0.995, compared to 0.81 and 0.78 in [3], 0.95 in [18], 0.962 in [16], and 0.887 in [22]. Additionally, the number of trainable parameters in [18] is 6,017,744 and in [16] is 9979000, which is very high compared to those of the proposed method, 2,155,436. Our method achieves a reduction of approximately 64.3% in the number of trainable parameters compared to [18] and 78.4% compared to [16]. This substantial decrease not only minimizes computational overhead and resource utilization but also contributes to potentially faster training and inference times. Such efficiency is particularly beneficial in real-time applications, like fall detection, where timely responses are critical. Table 5, Figs 13 and 14 present a comparison between the proposed model and state-of-the-art methods. It shows that the improved YOLOv8 is superior at detecting human fall events. Therefore, it offers greater reliability and assurance to the caregivers. The enhanced performance underscores the efficacy and efficiency of the proposed model. It underscores its potential for practical deployment in fall detection among the elderly. Moreover, the proposd model presents a good choice in terms of accuracy and speed when compared with EfficientDet, with highly configurable, allowing for the customization of hyper-parameters to further



FIGURE 15. Fall detection dataset.

optimize its performance. YOLOv8 is the recommended architecture for accurate and fast object detection with easy customization. However, the practical deployment and integration of YOLOv8 in real-time applications and multi-camera systems [48] involve selecting the appropriate model variant, training it on custom data, and optimizing it for the target platform. Next, the model is integrated into systems like autonomous vehicles, surveillance, retail, and agriculture to perform object detection tasks in real time [45], [46]. However, challenges include ensuring data privacy, managing computational resources, meeting latency requirements, and addressing potential model degradation [44]. Furthermore, maintaining high accuracy in diverse real-world conditions, such as varying lighting, occlusion, and object scale, is critical. On the other hand, resource constraints in embedded systems can limit model complexity and performance. Preserving data privacy and security while managing sensitive information is of utmost importance [49]. Besides that, it is essential to consider and counteract potential adversarial attacks that are specifically designed to deceive the model in order to ensure the reliable functioning of the system. It should be noted that the environment of the scene can also consider as a huge challenge in the detection, as well as the illumination. As such, all these challenge can make an obstacle to the practical deployment and integration of YOLOv8 in real-time applications.

VI. CONCLUSION

From this research, it is evident that the proposed fall detection method is capable of monitoring, identifying, and preventing human fall events with enhanced accuracy and speed. The advancements, including reducing the number

of layers and incorporating an attention mechanism into the YOLOv8 algorithm, have proven effective in detecting and tracking falls in real time. Therefore, the improved YOLOv8 model achieves a high detection accuracy of 99.5% and operates faster than previous YOLO algorithms. On comparing the proposed method with previous works for human fall detection in the elderly, the superiority of the proposed method is established. It demonstrates the model's practical effectiveness, reinforcing its suitability for real-world applications. This paper highlights crucial advancements in healthcare technology, emphasizing the need for reliable and efficient systems to detect falls among the elderly. These improvements not only enhance safety but also contribute to better care and prevention strategies in residential, occupational, and public places. However, there are certain limitations of this work. Variations in lighting, such as shadows, reflections, and changes in ambient light, can affect the model's ability to accurately detect objects. Poor lighting can obscure features and lead to incorrect detections. Besides, different environments can introduce a variety of obstacles and distractions. For instance, cluttered backgrounds, dynamic scenes with moving objects, and weather conditions like rain or fog can complicate the detection process. Moreover, low-resolution video and other camera-related problems, such as motion blur or lens distortions, can degrade the quality of the input data. This makes it harder for YOLOv8 to extract meaningful features and accurately identify objects. Therefore, several directions for future work can be suggested. Firstly, developing advanced fall detection systems capable of automatically triggering emergency alerts. Such systems could notify caregivers, family members, or emergency services, ensuring immediate assistance to individuals who have fallen. Another critical area for future research is optimizing the model further. By fine-tuning model parameters and adjusting the structure, researchers can improve detection accuracy while simultaneously reducing computational requirements.

This would make the model more efficient and suitable for deployment in resource-constrained environments, such as mobile devices or edge computing systems. Lastly, extending the system's detection capabilities to include a broader range of objects is another valuable direction for future work. While the current focus is on fall detection, incorporating additional object classes can significantly enhance the system's versatility and applicability.

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